L M M J V S D

FEBRERO /_____/

Ejercicio 04

$n = 463$

Intervalo	N. Empleados (n_i)	N_i (Frecuencia acumulada absoluta)	N_q
200 - 299	85	85	
300 - 399	90	175	
400 - 499	120	295	
500 - 599	70	365	
600 - 699	62	427	
700 - 799	36	463	

$C = 99$

$n = 463$

Cuando n:

impar	par
Cuartiles: $\frac{k(n+1)}{4}$	$\frac{k(n)}{4}$
Deciles $\frac{k(n+1)}{10}$	$\frac{k(n)}{10}$
Percentiles $\frac{k(n+1)}{100}$	$\frac{k(n)}{100}$

* Calcular: Q_1, Q_2, Q_3

D_6, D_7

P_{30}, P_{90}

$L_i = 300$

* Esta en el intervalo 2

$$Q_1 = \frac{1 \cdot 463}{4} = 115.75$$

* posición 2 → Intervalo 300-399

$$Q_1 = 300 + \left(\frac{115.75 - 85}{90}\right) \cdot 99$$

$$Q_1 = 300 + \left(\frac{30.75}{90}\right) \cdot 99$$

$$Q_1 = 300 + 0.3417 * 99$$

$$Q_1 = \boxed{333.83}$$

$$Q_2 = \text{posición} = \frac{2 * 463}{4} = 231.5 \; [400, 499]$$

$Fa = 175$ $\quad Li = 400$

$Fi = 120$

$$Q_2 = 400 + \left(\frac{231.5 - 175}{120}\right) \cdot 99$$

$$Q_2 = 400 + \left(\frac{56.5}{120}\right) \cdot 99 = Q_2 = 400 + 0.4708 \cdot 90$$

$$Q_2 = 400 + 46.61$$

$$\boxed{Q_2 = 446.61}$$

$$Q_3 = \text{posición} = \frac{3 * 463}{4} = 347.25 \; [500, 599]$$

$Fa = 295$

$Fi = 70$

$Li = 500$

$$Q_3 = 500 + \left(\frac{347.25 - 295}{70}\right) \cdot 99$$

$$Q_3 = 500 + \left(\frac{52.25}{70}\right) \cdot 99$$

$$Q_3 = 500 + 0.7464 \cdot 99 = 500 + 73.89 = \boxed{573.89}$$

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[400, 499]

$D_6 = \frac{6 \cdot 463}{10} = 277.8 \rightarrow L_i = 400,\ F_i = 120,\ F_a = 175$

[500, 599]

$D_7 = \frac{7 \cdot 463}{10} = 324.1 \rightarrow L_i = 500,\ F_i = 70,\ F_a = 295$

$D_6 = 400 + \left(\frac{277.8 - 175}{120}\right) \cdot 99$

$D_6 = 400 + \left(\frac{102.8}{120}\right) \cdot 99 = 400 + 0.8567 \cdot 99$

$D_6 = 400 + 84.81 = \boxed{484.81}$

$D_7 = 500 + \left(\frac{324.1 - 295}{70}\right) \cdot 99 = 500 + \left(\frac{29.1}{120}\right) \cdot 99$

$D_7 = 500 + 0.4157 \cdot 99 = 500 + 41.15 = \boxed{541.15}$

Cuantil 90:

$P_{90} = \frac{90 \cdot 463}{100} = 416.7$ [600, -699)

$L_i = 600,\ F_i = 62,\ F_a = 365$

$P_{90} = 600 + \left(\frac{416.7 - 365}{62}\right) \cdot 99 = 600 + \left(\frac{51.7}{62}\right) \cdot 90$

$P_{90} = 600 + 0.8339 \cdot 99 = 600 + 82.56 = \boxed{682.56}$

① Ejercicio Datos no agrupados.

3, 4, 4, 4, 5, 5, 6, 6, 6, 7, 7, 7, 7, 8, 8, 8, 8, 9, 9, 9, 9, 9,
10, 10, 10, 11, 11, 12, 12, 12, 12, 12, 13, 13, 14, 14, 14,
14, 14, 15, 15, 16, 16, 16, 17, 18, 18, 18, 19, 20, 20, 21,
21, 21, 22, 23, 24, 25, 25, 25, 26, 26, 26, 27, 27,
28, 30, 30, 31, 31, 32

Posiciones de los Cuartiles: $N = 80$

$i_1 = \frac{1(n+1)}{4} = \frac{81}{4} = 20.25$ valor entre posiciones 20(9) 21(9)

$Q_1 = \frac{9+9}{2} = 9$

$i_2 = \frac{2(n+1)}{4} = \frac{(81)2}{4} = \frac{162}{4} = 40.5$ posi.

Valor: 40 y 41
15 16

$Q_2 = \frac{15+16}{2} = \frac{31}{2} = 15.5$

$i_3 = \frac{3(80+1)}{4} = \frac{243}{4} = 60.75$

posiciones: 60 y 61
valores: 26, 26

$Q_3 = \frac{26+26}{2} = \frac{52}{2} = 26$

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NOTAS

Deciles 3, 5 y 7:

$n = 80$

$n+1 = 81$

$$D_k = \frac{k(n+1)}{10}$$

Posiciones:

$$D_3 = \frac{3(81)}{10} = 24.3 \longrightarrow$$ valor $24 = 11$, $25 = 11$

$$D_5 = \frac{5(81)}{10} = 40.5 \longrightarrow$$ valores: $40 = 15$, $41 = 16$

$$D_7 = \frac{5(81)}{10} = 56.7 \longrightarrow$$ valores: $57 = 25$, $56 = 25$

$$D_3 = \frac{11+11}{2} = \frac{22}{2} = 11$$

$$D_5 = \frac{15+16}{2} = 15.5$$

$$D_7 = \frac{25+25}{2} = \frac{50}{2} = 25$$

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NOTAS

Percentiles 34 y 80:

P_K = valor en la posición $\frac{K(n+1)}{100}$

$n=80 \quad n+1=81$

posiciones:

$P_{34} = \frac{34(81)}{100} = 27.54 \rightarrow$ posiciones (valores)
27 → 12
28 → 12

$P_{80} = \frac{80(81)}{100} = 64.8 \rightarrow$ posiciones
64 → 27
65 → 28

$P_{34} = \frac{12+12}{2} = \frac{24}{2} = \boxed{12}$

$P_{80} = \frac{27+28}{2} = \frac{55}{2} = \boxed{27.5}$

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